

Lab Report: Physics-Informed Neural Networks

General Guidelines

- Language: English
- Use plots created during the course
- Focus on content, not on implementation details
- For key results, follow this evaluation structure:
 1. What did we do?
 2. What do we see in the plot?
 3. How do we interpret what we see?
 4. Evaluation of our interpretation
- Submission Deadline: No later than 2 weeks after the course ends

Theory Section

- Length Estimate:
 - ~5 pages (single column)
 - ~3 pages (double column)
- Focus: Concise explanation of key ideas
- Topics to include:
 - Neural Network (NN) architecture (mainly MLPs)
 - NN training process
 - Introduction to PINNs and how they differ from regular NNs

Day 1

System of Investigation:

- Describe the physical system
- Define the learning task

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Investigating Hyperparameters:

- Explore optimal learning rate
- Discuss why a sweet spot exists

NN vs. PINN:

- Compare performance of standard NN vs. PINN
- Analyze observed differences
- Consider extrapolation: Does the PINN generalize beyond training data?

Varying Collocation Points:

- How many collocation points are needed with only 2 data points?
- Why does the model still find the correct solution despite the underdetermined setup?

Day 2

System of Investigation:

- Describe the physical system
- Present the dataset
- Explain the learning task
 - Include equations for:
 - Loss functions
 - Residuals
- Clarify the role of each loss term

Learning Time-Dependent Wave Functions:

- Show loss evolution during training
- Visualize wave functions over time
- Explain why this behavior is expected
- Compare to analytical computation (how would we solve this without a PINN?)

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Energy Eigenvalue:

- Derive how the energy eigenvalue is computed
- Present relevant plots
- Discuss:
 - Why the energy eigenvalue has an imaginary part (should be real by definition?)
 - Its evolution during training

Diminishing Potential:

- Show potential (both image and equation)
- Visualize the resulting wave functions
- Explain:
 - Why time-dependent potentials are hard to solve analytically
 - Why PINNs handle time dependence naturally

Summary

- Recap key insights from Day 1 and Day 2
- Reflect on why PINNs are well-suited for physics problems